Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

***Benchmark:** LAILA | **Context:** Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

This is the dimension with the most direct deployment mismatch and is flagged HIGH priority in elicitation. LAILA's evaluation produces only discrete point-score labels [Q181, Q182] evaluated by Quadratic Weighted Kappa [Q58]. The benchmark does not evaluate confidence calibration, uncertainty quantification, score-range prediction, or rationale generation. The deployment use case explicitly requires confidence signals and explanatory justification [elicitation Q3] so teachers can intervene meaningfully on borderline essays. There is no overlap between the benchmark's evaluated output form and the deployment's required output form for these features. Web research confirms this is a structural gap in the field: rationale-generation frameworks (RMTS, RaDME, QwenScore+) exist for English but no Arabic adaptation has been documented [WEB-12, WEB-13, WEB-14], and TAQEEM 2025 used Chain-of-Thought prompting but focused on accuracy, not rationale [WEB-15]. The benchmark's own ethics statement explicitly prohibits use 'for high-stakes educational decisions affecting individual students' [Q104] and warns against 'deployment in operational assessment contexts without independent validation' [Q105]. Combined with the elicitation marking both OO and OF as HIGH priority for this exact reason, the score is 1 — fundamental misalignment.

ID	Evidence
Q3	“Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings.” (p.1)
Q58	“We evaluate model performance using QWK to measure agreement between human and model scores.” (p.8)
Q104	“We explicitly prohibit its use for high-stakes educational decisions affecting individual students.” (p.10)
Q105	“Users must commit to: (1) preventing re-identification attempts, (2) avoiding demographic profiling or inference, and (3) refraining from deployment in operational assessment contexts without independent validation.” (p.10)
Q181	“In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics.” (p.18)
Q182	“The holistic score was computed as the sum of the individual trait scores.” (p.18)
WEB-12	BERT-based Arabic AES system documented but without rationale generation (https://www.researchgate.net/publication/382302599_Automated_essay_scoring_in_Arabic_a_dataset_and_analysis_of_a_BERT-based_system)
WEB-13	RaDME generates trait scores with co-produced rationales using knowledge distillation — English only, no Arabic adaptation documented (https://arxiv.org/html/2502.20748v1)
WEB-14	QwenScore+ uses rubric-aware Chain-of-Thought prompting with RLHF — English only (https://www.preprints.org/manuscript/202504.2338)
WEB-15	TAQEEM 2025 used CoT prompting on Arabic essays but focused on scoring accuracy, not rationale output (https://aclanthology.org/2025.arabicnlp-sharedtasks.135.pdf)

Strengths

- QWK as the evaluation metric matches the metric used for IAA, providing internal coherence between human and model agreement [Q44, Q58]
- Three-seed repetition for LLM experiments supports reporting variability [Q189]
- Structured JSON output enforced via outlines library [Q190] could be extended to include rationale fields

ID	Evidence

Q44	“We compute IAA using Quadratic Weighted Kappa (QWK) (Williamson et al., 2012) to assess agreement between the two main annotators (A1 and A2), prior to adjudication. Table 4 reports QWK per trait and prompt, with agreement strength following Landis and Koch (1977). Overall, IAA was substantial across prompts, with an average QWK ranging from 0.66 (P5) to 0.75 (P1), showing strong consistency.” (p.6)
Q58	“We evaluate model performance using QWK to measure agreement between human and model scores.” (p.8)
Q189	“To account for variability in example selection, each experiment was repeated three times with random seeds 1, 11, and 42, and we report the average performance across runs.” (p.18)
Q190	“For all the LLM experiments, we used vLLM for inference with the outlines library to enforce the JSON output format.” (p.18)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: No. Benchmark output is discrete trait scores plus holistic sum [Q181, Q182]; deployment requires per-trait scores plus confidence signal plus rationale text. The benchmark does not evaluate the latter two.

OF-2: Not applicable — neither benchmark nor deployment requires speech output. Output is text only on both sides.

OF-3: INSUFFICIENT DOCUMENTATION on this specific point in the LAILA paper, but the deployment population is qualified teachers; literacy/accessibility considerations apply more to student-facing outputs than to the teacher-facing scoring interface evaluated here.

OF-4: Documented form mismatches harming external validity: (1) no confidence/uncertainty output evaluated [Q181]; (2) no rationale output evaluated [Q181, Q182]; (3) no in-domain Arabic precedent for rationale-generation evaluation [WEB-12, WEB-13, WEB-15]; (4) benchmark explicitly prohibits high-stakes operational use without independent validation [Q104, Q105].

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Q104	“We explicitly prohibit its use for high-stakes educational decisions affecting individual students.” (p.10)
Q105	“Users must commit to: (1) preventing re-identification attempts, (2) avoiding demographic profiling or inference, and (3) refraining from deployment in operational assessment contexts without independent validation.” (p.10)
Q181	“In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics.” (p.18)
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Information Gaps

- Whether any Arabic-language AES system has been validated with teacher-facing rationale output (gap_id 3: not found)
- Whether Arabic teachers would accept LLM-generated rationales as professionally credible

Requires Expert Verification

- Acceptance of AI-generated trait scores and rationales by Arab-country secondary Arabic teachers across the deployment range

Remediation

Gap 1: Benchmark evaluates only discrete point-score outputs via QWK; deployment requires uncertainty signals and rationale text [Q58, Q181]

Recommendation: Define and report new metrics alongside QWK: (1) calibration metrics for confidence outputs (e.g., expected calibration error against teacher disagreement); (2) rationale quality metrics evaluated by qualified Arab teachers across multiple countries. Treat current QWK results as necessary but not sufficient evidence.

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Q58	“We evaluate model performance using QWK to measure agreement between human and model scores.” (p.8)
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